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European Patent Office
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Munich
Germany

07 October 2022

Our Ref: NODT-004/N01EP 330150-2065

Dear European Patent Office

Re. European patent application no. 18826609.2
SULPHONYL UREA DERIVATIVES AS NLRP3 INFLAMMASOME MODULATORS
Nodthera Limited

We refer to the communication pursuant to Article 94(3) EPC dated 29 March 2022 and to the above-identified application.

We lodge herewith an amended set of claims to be substituted for the claims as presently on file. For the convenience of the Examiner, a further copy of the amended claims showing the amendments in track-changes format is also enclosed.

The applicant respectfully requests that any amendment of the description be deferred until such time as an allowable claim set has been agreed upon.

Amendments

The preamble of claim 2 has been amended to refer to claim 1.

Claim 9 has been amended to insert the structures of compounds nos. 1-166. This amendment finds basis in original claim 88 and Table 1 of the application as filed.

Clarity

The Examiner has objected that claim 2 as on file makes reference to “any of the preceding claims”. As set out above, claim 2 has been amended to refer to claim 1 only. As such, this objection is no longer relevant.

The Examiner has objected that claim 9 as on file makes reference to compounds disclosed in the description. As set out above, claim 9 has been amended to insert the structures of these compounds. Accordingly, this objection is no longer relevant.

Novelty and inventive step

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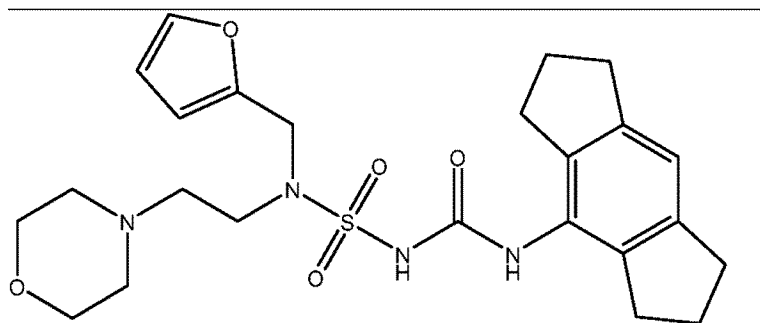
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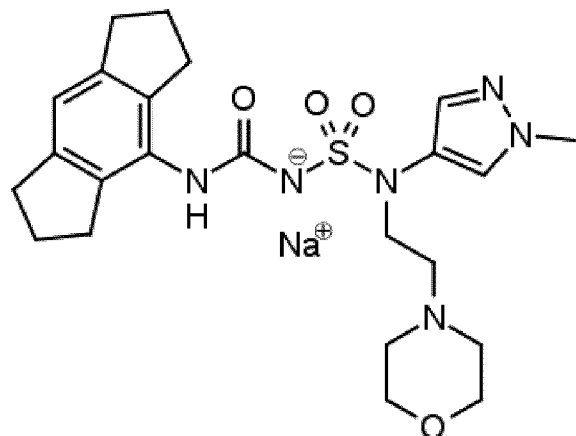
We are pleased to note that the Examiner has recognised that the claims on file are novel. The Examiner has, however, objected that the subject matter of the claims on file lacks an inventive step in view of D1. The Examiner has particularly highlighted Examples 17, 35, and 39 of D1.

In view of the fact that the Examiner has particularly highlighted Example 17 from D1 (see Sheet 4, paragraph 1 of the Official Communication which states “*in particular in view of example 17 of D1*”), the Applicant has now synthesised and tested the compound of Example 17 from D1 in the PBMC assay described in the present application as filed (see pages 294 to 295 of the application as filed) and the majority of tests returned an IC50 value of $\geq 10 \mu\text{M}$.

The compound of Example 17 has the following structure:



By contrast, from a review of all compounds within the present application, we submit that the closest compound (which shares the most structural features) to Example 17 from D1 is Example 46:



Example 46 of the present application differs from Example 17 of D1 by virtue of containing a 1-methyl-1H-pyrazol-4-yl group in place of the -CH₂-furan-2-yl group present in Example 17 of D1.

It can be seen that the corresponding PBMC IC50 of Example 46 of the present application returned a value of “++++” (see Table A on page 297 of the application as filed) which corresponds to an IC50 value of $<1 \mu\text{M}$ (see page 295 of the application as filed).

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Therefore, the effect of such a substitution is greater than 10 fold (i.e. from $< 1 \mu\text{M}$ to $> 10 \mu\text{M}$).

We submit that this comparative data clearly demonstrates that the skilled person would be surprised that such a substitution would have such a profound and measurable effect on the potency of the resultant compounds (i.e. greater than a 10 fold increase in potency).

Even if the Examiner is minded not to agree that this comparative data is compelling evidence of the presence of an inventive step, we also submit that the present case is inventive due to structural non-obviousness over D1. For example, there are only 8 compounds in D1 which have bis-substituted nitrogen atoms (see Examples 3, 17-20, 29 and 31-32 of D1). None of these compounds in D1 contain a heteroaryl ring directly attached to the nitrogen atom as required by present claim 1. By contrast, present claim 1 requires the corresponding nitrogen atom to be bis-substituted (i.e. neither R^2 nor R^3 can be hydrogen) and the R^3 substituent (the group which corresponds to R^1 in D1) must be directly attached. Thus, we submit that the skilled person would not have been motivated to remove the linker between the nitrogen atom and the heteroaryl ring in D1 based on the fact that it is present in all cases.

Accordingly, claim 1 involves an inventive step in view of D1 at least by virtue of the above reasons. Claims 2-15 either depend upon claim 1 or incorporate its subject matter by reference and therefore likewise involve an inventive step at least by virtue thereof.

The Applicant submits that this response overcomes all objections raised in the Examination Report. If, however, the Examiner disagrees, they are requested to issue another Official Communication, or to contact the undersigned for any minor outstanding matters. Purely as a precaution, should the Examiner be minded to refuse this application, oral proceedings via videoconference are requested under Article 116 EPC before any such decision is taken.

Yours faithfully



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For and on behalf of Cooley (UK) LLP
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